

Design and Analysis of MAC Unit Using Single Precision Floating Point Vedic Multiplier

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Abstract

Multiplication and accumulation are the basic operations which are important in several microprocessors and digital signal processing (DSP) applications to execute dedicated algorithms. Developing high speed MAC is essential for real time DSP application. The MAC unit determines the speed of the overall system and it lies in the critical path. As the demand for high speed design is continuously increasing, the studies related to the field of multipliers and adders are endless and still significant. Multiplication of floating point numbers gives high resolution and has found extensive use in DSP applications. The critical part in single precision floating point multiplication is the 24×24 bit mantissa calculation. The speed of the system can be improved by enhancing the speed of multiplication. In this study, a 24 bit Vedic multiplier has been proposed using 3×3 Vedic multiplier as its basic block. This study proposes MAC unit using modified single precision floating point Vedic multiplier using 'Urdhava-Triyakbhyam' sutra. In this work, MAC unit for DSP application using 32 bit modified single precision floating point Vedic multiplier is simulated in Xilinx ISE 14.7 using Virtex 6 FPGA.

Keywords: Multiply and accumulate, Institute of Electrical and Electronics Engineers, digital signal processing

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INTRODUCTION

The DSP functions extensively make use of multiply and accumulate operation for high performance. In most of the digital signal processing applications, the critical operations are the multiplication and accumulation; so, the research focuses on MAC design to enhance its speed. In real time signal processing applications, high throughput and high speed MAC unit with low power consumption is always a key to achieve a high performance in DSP applications. So, developing a high speed and high throughput MAC unit is essential for real time DSP application. Moreover, with the ever increasing demand for portable electronic devices, an electronic component with low power consumption, high speed and area efficiency would surely lead the market trend. Therefore, designing MAC architecture with high computational performance would be promising. The main goal of this study is

designing of MAC unit using single precision floating point Vedic multiplier.

The increased use of portable and handheld devices paved the way for need of further advancements in the area of VLSI design. Due to the increased demand, low cost and power efficient design constraints should be satisfied. Thus the studies related to the field of multipliers and adders are endless and are still important. Many ideas from device level to architectural levels and above have already been proposed. However, there is no universal method to avoid trade-offs between power, delay and area. So, the technique chosen by designers must satisfy their application and needs. Most of the portable electronic devices require digital signal processing for high performance. Multiplication as one of the most complex arithmetic operations are widely used in various applications of DSP. Since multiplication requires more number of steps

for the processing and design, and it dominates the execution time of most DSP algorithms, so there is a need of high speed multiplier. Different topologies are there for fast multiplication. The multiplier used inside the MAC unit is based on the sutra “Urdhva Tiryagbhyam” which is one of the multiplication formulas of Vedic mathematics.

There are many data processing applications like image and voice processing which use a large number of values and need a relatively high precision, or high resolution, a relocatable radix point is also desirable. In such cases, instead of encoding the information in the form of integers or fixed point numbers, an alternative solution is a floating point representation. In a fixed point representation, it places a limit on the magnitude and precision of the number representation. Floating point numbers are one of the possible ways of representing real numbers in binary format. The advantage of floating-point numbers is that they can represent very large and very small values. The name floating point comes from the fact that the radix point is floating.

In this study, floating point numbers are represented according to IEEE 754 standard format. IEEE 754 standard is a technical standard established by IEEE and most frequently used standard for floating point computation, and is followed by many CPU and FPU implementations. The standard defines floating point formats; single precision consist of 32 bit and double precision consist of 64 bits.

Xilinx Virtex VI platform FPGA usage for the implementation of floating point numbers instead of microprocessor based configurations will be the leading choice due to high speed operation, parallel processing and re-programmability. This study describes the design of MAC using single precision floating point multiplier based on Vedic mathematics. Proposed MAC will be modeled using Verilog. The design is synthesized and simulated using ISE simulator.

LITERATURE SURVEY

Paldurai and Hariharan designed a FPGA implementation of delay optimized single precision floating point multiplier which

shows an improvement in multiplication speed by 21.7% when compared to conventional multiplier [1]. Optimization of the speed and reduction in area of the multiplier is a major concern while designing a circuit. This study shows that by reducing gate delays, the overall performance of digital circuits can be improved. The lesser number of LUTs verifies that the hardware requirement is reduced; thereby the power consumption is also reduced. The time required for computing multiplication is less than the other multiplication techniques.

This study describes the implementation of pipelining in the design of floating-point multiplier. Pipelining is one of the popular methods to realize high performance computing platform and as the number of pipelining stages increases, speed and throughput also increase. Pipelining is a technique where multiple instruction executions are overlapped in order to maximize the speed of the multiplier. Padma *et al.* proposed a pipelined floating point multiplier based on Vedic multiplication technique where the LUTs used are less in number which implies reduced hardware requirement, thereby reducing the power consumption [2]. Teja *et al.* designed a FPGA implementation of low-area floating point multiplier using ‘Vedic Nikhilam Sutra’ [3]. Here the area has been reduced because of the less usage of CLBs and flip flops. This study describes that purpose of using Vedic mathematics is due to increase in the number of partial products in normal multiplication process; by using Vedic mathematics, partial products can be reduced so that the area and power constraints of the floating point multiplier can be reduced efficiently.

Koneru *et al.* proposed asynchronous single precision floating point multiplier using Verilog HDL [4]. Here, the mantissa part of floating point format is done using Booth’s algorithm and Vedic multiplier where they showed that the Vedic multiplier was surpassing in terms of path delay and area. Vaibhav *et al.* designed a VHDL implementation of floating point multiplier using Vedic mathematics to obtain the improvement in power efficiency; the ‘Urdhva

Tiryagbhyam' is implemented and they found that the components can be decreased which in turn reduces the complexity of the hardware circuit [5]. Ravi *et al.* proposed an analysis and study of different multipliers to design floating point MAC units for digital signal processing applications [6]. They showed here that the floating point multiplier can be run at the frequency 80 MHz when implemented on FPGA.

The study by Jain *et al.* describes MAC unit which performs both multiplication and addition functions [7]. It operates in two stages: Firstly it computes the product of given numbers and forwards the result for the second stage operation i.e. addition/accumulate. The performance of the MAC can be increased by optimizing the basic blocks that are used in the design of the overall MAC unit.

ARCHITECTURE OF THE SYSTEM

Floating Point Number

Floating point number means *that* the radix point can be shifted to right or left of the fixed number, hence the name floating point. The floating point representation gives more precision and they can represent a much larger range of values compared to the fixed point representation.

Floating Point Representation

In single precision, a floating point number consists of a 32 bit word divided in 1 bit for sign, 8 bits for exponent that constitutes a 127 bias, and 23 bits for the mantissa (Figure 1).

Sign	Exponent	Mantissa
31	30 23 22	0

Fig. 1: Single Precision Format.

The special case numbers as defined by the IEEE-754 standard and their data bit representations are shown in Table 1.

Vedic Mathematics

Vedic mathematics is used to solve complex calculations into simple one. Vedic mathematics was reorganized from the ancient Indian scriptures (Vedas) by Swami Bharati Krishna Tirthaji Maharaja (1884–1960) after his eight years of studies related to Veda; he came up with 16 sutras and 13 sub-sutras which deal with various branches of

mathematics like arithmetic, algebra, geometry, factorization etc. Out of these sutras, 14th sutra which is 'Urdhva Tiryagbhyam' is used in the present study. 'Urdhva Tiryagbhyam' exhibits high degree of parallelism and employs parallel multiplication compared to other sutras. A demonstration of two-decimal number multiplication using 'Urdhva Tiryagbhyam sutra' is shown in Figure 2.

Table 1: Special Case Numbers in IEEE-754 Representation.

Meaning	Sign Field	Exponent Field	Mantissa Field
Zero	Do not care	All 0s	All 0s
Positive Denormal	0	All 0s	Non-Zero
Negative Denormal	1	All 0s	Non-Zero
Positive Infinity	0	All 1s	All 0s
Negative infinity	1	All 1s	All 0s
Not a Number (NaN)	Do not care	All 1s	Non-Zero

Step 5	Step 4	Step 3	Step 2	Step 1
1 0 1	1 0 1	1 0 1	1 0 1	1 0 1
	×	×	×	
1 1 0	1 1 0	1 1 0	1 1 0	1 1 0
1	1	1	1	0

Fig. 2: Urdhva Tiryakbhyam Sutra for 3×3 Block.

FLOATING POINT MULTIPLICATION

Multiplication of two floating point numbers represented in IEEE 754 format is shown below. Figure 3 shows the multiplier structure that includes the addition of exponents, multiplication of mantissa, and sign calculation. This structure remains same for all precision floating point multipliers according to the standard of IEEE 754. Multiplication of two floating point binary digits represented in IEEE 754 general format is interpreted as:

$$R = (-1)^{\text{Sign}} \times 2^{(\text{exponent} - \text{bias})} \times 1.\text{fraction}$$

Binary Floating Point Multiplication Steps

Binary floating point multiplication is done in four steps. They are explained below:

Mantissa Calculation Unit

Multiplying the significant: mantissa output = 1.F1 × 1.F2

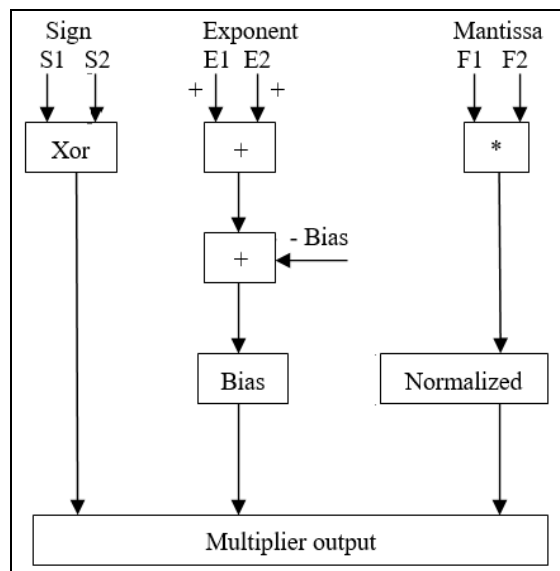


Fig. 3: Floating Point Multiplier Structure.

Take the 23 bit mantissa part; always add 1 at the MSB, so now it has 24 bits. Multiply these two 24 bit mantissa by using the proposed Vedic multiplier and get the 48 bit result.

Exponent Calculation Unit

As the E1 and E2 exponent fields are biased, the bias must be removed in order to do the addition. And then, we must add again the bias, to get the value to be entered into the exponent result.

$$\text{Exponent output} = (E1 - 127) + (E2 - 127) + 127 = E1 + E2 - 127.$$

Here three 8 bit numbers are added in order to calculate the exponent by using carry look ahead adder, because carry look adder has better performance in terms of speed.

Sign Calculation Unit

$$\text{Sign output} = s1 \text{ XOR } s2$$

If S is 0, it denotes the given value is positive, and if S is 1, it denotes the value as a negative number.

Normalization

The result of the mantissa multiplication must be normalized to have a leading '1' just to the left of the radix point. If the leading one is at bit 46 of mantissa output, then the intermediate product is already a normalized number. If it is available at bit 47 of mantissa output, then the intermediate product is shifted to the right and increment the exponent calculation unit output by 1.

PROPOSED SYSTEM

The proposed system aims to improve speed using Vedic multiplier. The main applications of Multiply-Accumulate (MAC) unit are microprocessors, logic units and digital signal processors, since it determines the speed of the overall system. Multiplier is a critical component to speed up the MAC unit. By using Vedic multiplier, critical delay can be decreased than by conventional multipliers. Vedic multiplier with 'Urdhava-triakybhyam' sutra for 24 bit multiplication is proposed. The proposed system is designed to improve speed of existing MAC. The hardware implementation of the basic 3×3 block is shown in the below Figure 4 and it is designed using Simulink.

The block diagram of the proposed system is shown in the Figure 5 below. For 24-bit multiplier design; first, the basic block 3×3 bit multiplier is designed, then a 6×6 block is designed using this 3×3 block, then a 12×12 block is designed using this 6×6 block, and finally a 24×24 bit block is designed using this 12×12 block. From these multipliers, it is concluded that to construct a 24×24 bit Vedic multiplier, four 12×12 Vedic multipliers and three carry look ahead adders are required. In this method, the partial products are generated parallel which itself reduces delay and makes this method fast.

The proposed system is designed to improve speed of existing MAC unit. MAC unit performs both multiply and addition functions. The input is being fed from the memory location. When the input is given to the multiplier, it starts computing value for the given input. If the input is n bit, then the output will be $2n$ bits. The multiplier output is given as the input to adder which performs addition. The output of adder is $2n+1$ bit, i.e., one bit is for the carry ($2n$ bits+1 bit). The floating point adder works if the exponents are same. After comparing the exponents of the two numbers, shift the smaller number to the right until its exponent matches the larger exponent. Compute the sum/difference of the mantissa depending on the sign bit. If signs of inputs are equal, then add the mantissa or if signs of inputs are not equal then subtract the mantissa. Normalize the result if needed.

Then, the adder output is given to the accumulator register. The output of the

accumulator register is taken out or fed back as one of the input to the adder (Figure 6).

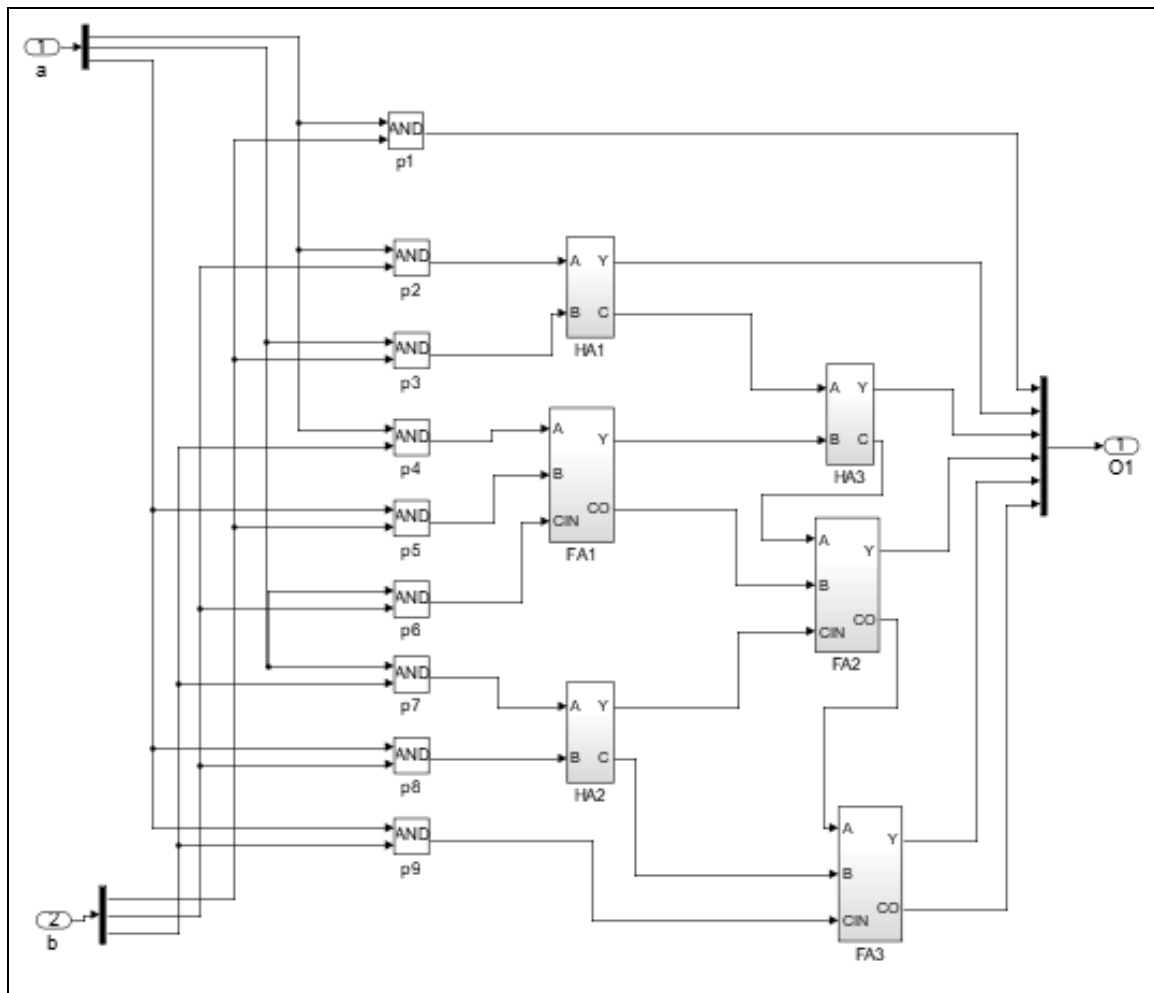


Fig. 4: Hardware Implementation of 3×3 Block.

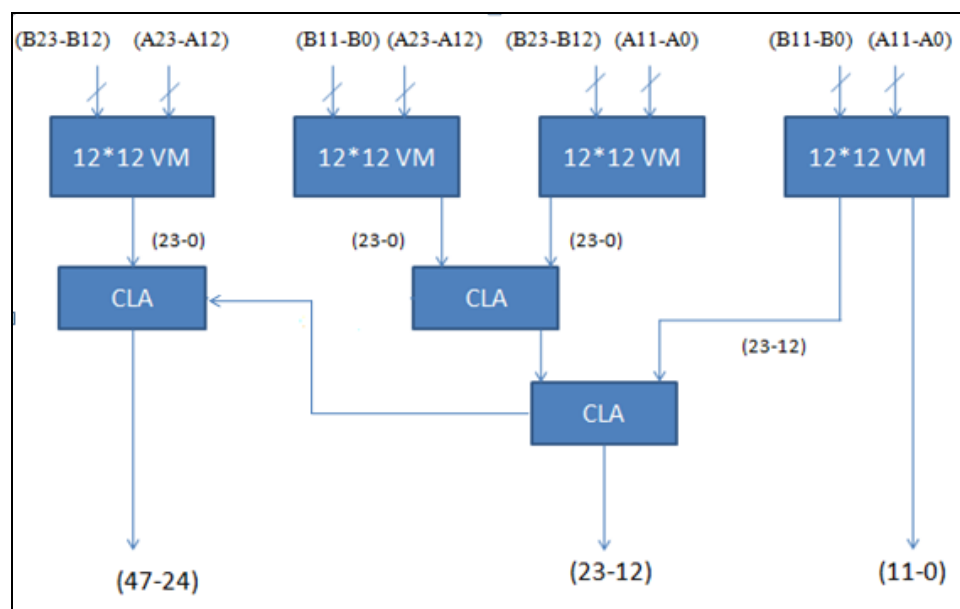


Fig. 5: Block Diagram of 24×24 Multiplier.

EXPERIMENTAL RESULTS

The design is developed using Verilog HDL and synthesized in Xilinx ISE 14.7 platform. The block diagram, RTL schematic and the simulation results of MAC using IEEE 754 single precision floating point Vedic multipliers are shown in Figures 7–10.

The simulation results of the 24×24 block Mantissa part of the IEEE 754 single precision floating point number ‘Urdhva-Tiryagbhyam’ Vedic multiplier is shown Figure 11. The device utilization summary and timing details of IEEE 754 Single precision floating point multiplier and MAC unit on Virtex VI FPGA are shown in Tables 2 and 3 respectively. The proposed Vedic multiplier and conventional Booth multiplier, array multiplier and Wallace multiplier are coded in Verilog, synthesized and simulated using ISE simulator. Then their comparison on the basis of delay is shown in Table 4.

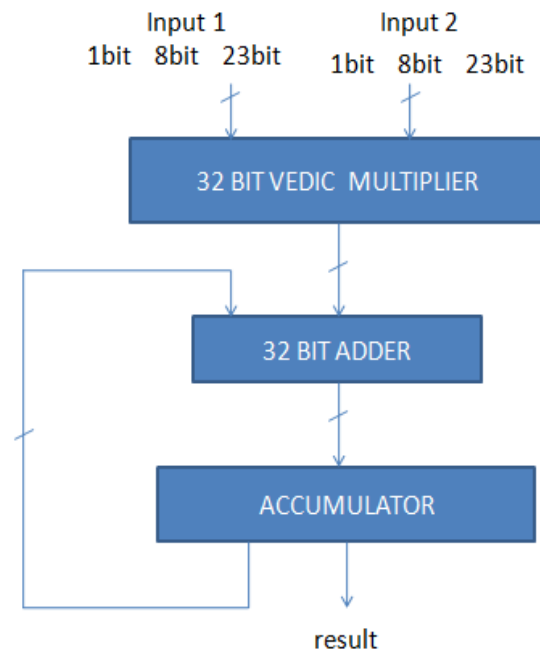


Fig. 6: Block Diagram of MAC Using Vedic Multiplier.

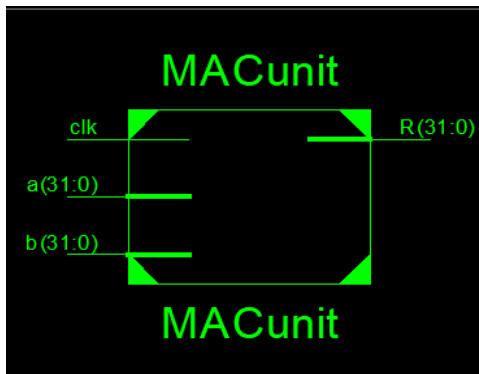


Fig. 7: Block Diagram of MAC Unit.

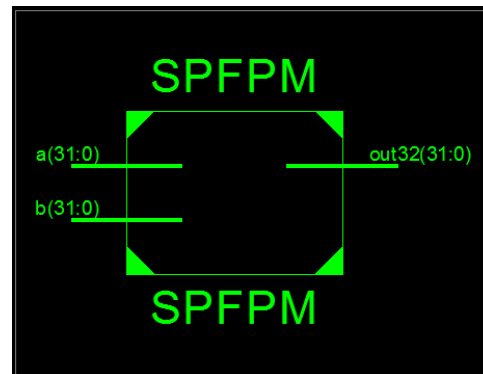


Fig. 8: Block Diagram of IEEE 754 Vedic Single Precision Floating Point Multiplier.

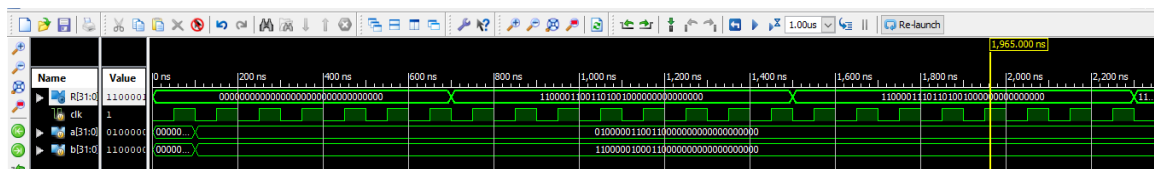


Fig. 9: Simulation Result of MAC Unit.

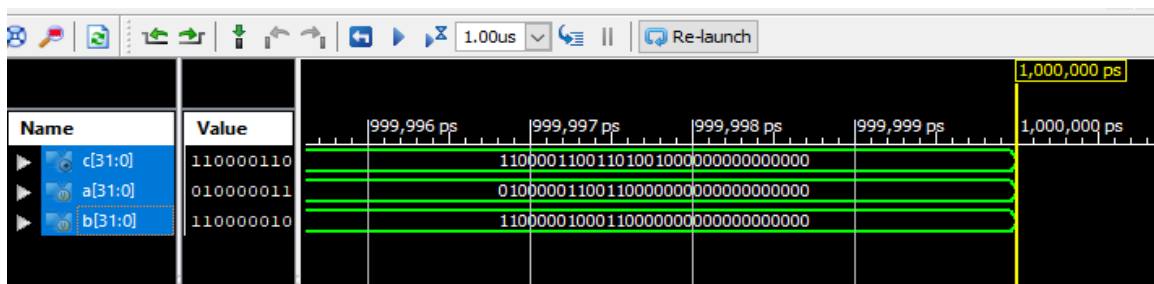


Fig. 10: Simulation Result of IEEE 754 Vedic Single Precision Floating Point Multiplier.

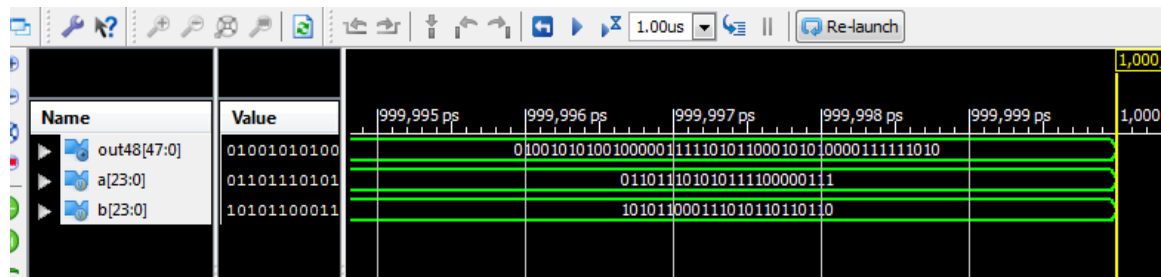


Fig. 11: Simulation Result of 24×24 Vedic Multiplier.

Table 2: Device Utilization Summary for SP.

No. of Slice LUTs Used	No. of Bonded IOBs Used	Delay
1241/465600 (2%)	128/240	17.889 ns

Table 3: Device Utilization Summary for MAC Unit.

No. of Slice LUTs Used	No. of Bonded IOBs Used	Delay
688/465600 (1%)	98/240	6.837 ns

Table 4: Delay Comparison of Modified Multiplier with Existing Multipliers.

Types of Multipliers	Delay (ns)
Array multiplier	36.43
Wallace multiplier	44.43
Booth multiplier	64.01
Vedic multiplier	18.01

CONCLUSION

This study proposes a MAC unit using floating point multiplier that supports the IEEE 754 single precision floating point standard. Vedic multiplication procedures are quite simple and efficient compared to the conventional multiplications. Multiplier is one of the most important blocks in MAC unit. The overall performance of the MAC unit depends on the resources used by the multiplier. Using Vedic multiplier, critical delay can be decreased than by conventional multipliers. Design of all DSP operations with the help of high speed Vedic multipliers increases the efficiency of system and reduces the processing time. Rapidly growing technology has raised demands for fast and efficient real time digital signal applications. 'Urdhava-triyakbhyam' sutra uses parallel generation of partial products and removes unwanted steps, and also, 'Urdhva Tiryagbhyam' Vedic multiplier gives the complete multiplication calculation in one line, hence occupies less space which implies reduction in delay. Also, hardware components used are less and operating speed is high due to crosswise and vertical

calculation; hence it reduces complexity. The number of LUTs and slices required for Vedic multiplier is less, due to which the power consumption is reduced.

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